BIG IDEAS MATH • GRADE 8 • CHAPTER 1

Solving Multi-Step Equations: Variables on Both Sides

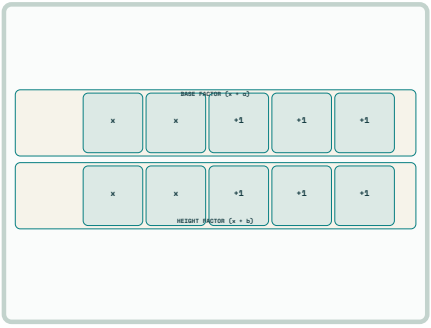
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.8.EE.C.7

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Algebra Stage

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Place 5x and 3 units on left tray; place 2x and 15 units on right tray. Remove 2x from both, then remove 3.

Sketch Model on Paper:
Draw balance beam with 5x + 3 on left and 2x + 15 on right.

3. Step-by-Step LCD Stack Trace (SC-32)

MODE: 123 Base-10 Arithmetic

1.	15	X: 15	Constant 15 into X
2.	ENTER	Y: 15 | X: 15	Stack lift
3.	3	Y: 15 | X: 3	Constant 3 into X
4.	-	X: 12	Delta constants: 15

EXPECTED LCD X: x = 4

1 **Solve 5x + 6 = 2x + 21 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

2 **Solve 7x - 5 = 3x + 19 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

3 **Solve 8x + 4 = 4x + 28 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

4 **Solve 9x - 12 = 4x + 18 on the stack.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

Station Diagnostic & Solution Key: Chapter 1

Solving Multi-Step Equations: Variables on Both Sides • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Tactile Algebra Stage & Tile Weights

CAD Blueprint: algebra_stage

Target Device: SC-32 Scientific

CAD / STL Source File:

Education/build/scad/algebra_stage.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Oceanic Teal Stage Frame & White Algebra Tiles (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Solve 5x + 6 = 2x + 21 on the stack.** Standard: CCSS.MATH.8.EE.C.7	x = 5	[2][1] [ENTER] [6] [-] [5] [ENTER] [2]
2	**Solve 7x - 5 = 3x + 19 on the stack.** Standard: CCSS.MATH.8.EE.C.7	x = 6	[1][9] [ENTER] [5] [+] [7] [ENTER] [3]
3	**Solve 8x + 4 = 4x + 28 on the stack.** Standard: CCSS.MATH.8.EE.C.7	x = 6	[2][8] [ENTER] [4] [-] [8] [ENTER] [4]
4	**Solve 9x - 12 = 4x + 18 on the stack.** Standard: CCSS.MATH.8.EE.C.7	x = 6	[1][8] [ENTER] [1][2] [+] [9] [ENTER]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Watch for combining terms on opposite sides without changing their sign.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.